COURSE SLIDES · WSQ

Agentic AI for Business Process Automation

WSQ Course Code: TGS-2024045801

TSC: Artificial Intelligence Application · AER-TEM-3026-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE
Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS
PhD — specialises in AI, automation and software engineering.

DELIVERS
WSQ courses on agentic AI, workflow automation (n8n) and app development.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

1

Who you are

Your name, your organisation and your role.

2

Your starting point

Your experience with automation or AI tools — including none at all.

3

Your pain point

One business process in your work you would most like to automate.

4

Your goal

What you want to be able to build by the end of this course.

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HOUSEKEEPING

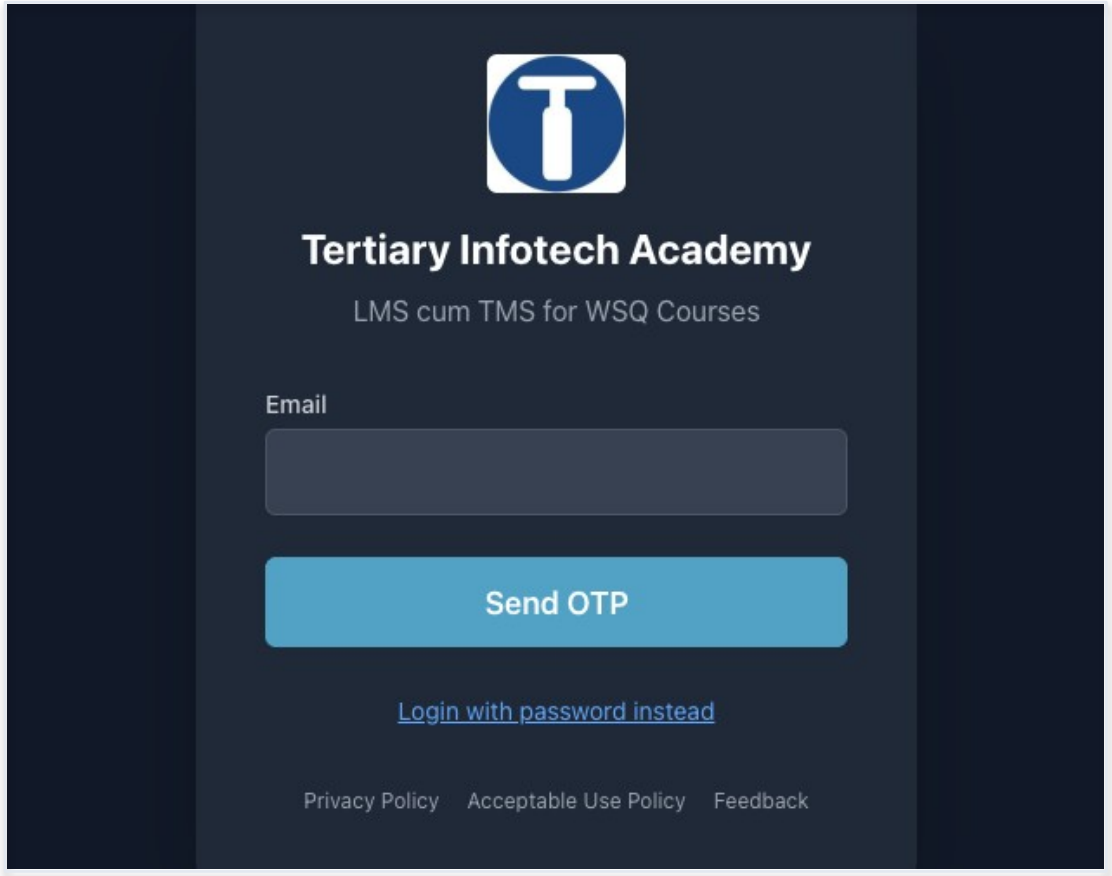
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Download Course Material



Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

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COURSE PORTAL · STEPS

Download Course Material

1. Go to lms-tms.tertiaryinfotech.com
2. Enter your registered email and click Send OTP
3. Key in the one-time password from your inbox
4. Open this course and download the slides + Learner Guide
5. Keep them open — the assessment is open book

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GET THE FLOWS

Get the Course Workflows

1

Every lab ships a workflow

A ready-made n8n workflow (.json) sits in each lab's folder under labs/.

2

Import it

In n8n: Workflows → Add workflow → the '...' menu → Import from File → pick the lab .json.

3

Re-select your credentials

After importing, re-select YOUR OWN credentials on every node showing a warning.

4

Nothing of yours is shared

Imported workflows reference credential NAMES, not secrets — no key travels in the file.

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WSQ ALIGNMENT

Skills Framework (TSC)

1

TSC Title

Artificial Intelligence Application

2

TSC Code

AER-TEM-3026-1.1

3

Mapped to the national framework

This WSQ course is mapped to the SkillsFuture Singapore Skills Framework.

4

Fully assessed

The assessment covers every Ability (A) and Knowledge (K) statement listed next.

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SKILLS FRAMEWORK

TSC Abilities and Knowledge

ABILITIES (assessed by the PP)

- A1 Implement AI applications in collaboration with technology service providers
- A2 Deploy AI workflows according to plan
- A3 Identify and report any issues with the AI applications and data collected

KNOWLEDGE (assessed by the WA)

- K1 Background and basics of AI
- K2 Fundamental concepts and methods of statistics and programming for data science
- K3 Mathematics and computing theories
- K4 Machine learning techniques and applications
- K5 Principles of data management
- K6 Software development methodologies
- K7 Types of AI applications
- K8 AI implementation procedures

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SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

DAY 1 — n8n, AI Agents and Agentic Chatbots

- Topic 1: Overview of n8n and AI Agents
- Lab 1 — n8n trigger and action
- Lab 2 — Webhook and AI Agent
- Topic 2 begins: agentic chatbots
- Lab 3 — Chatbot webhook
- Timing: 9:30 am – 6:30 pm

DAY 2 — Human in the Loop, RAG and Assessment

- Topic 2: Human in the loop
- Lab 4 — Human in the loop
- Topic 3: RAG
- Lab 5 — RAG customer care chatbot
- Final Assessment (WA + PP)
- Timing: 9:30 am – 6:30 pm instruction, assessment to 8:45 pm

Each day is 8 instructional hours: a 1-hour lunch is excluded, and the two 15-minute tea breaks are counted within the instructional time

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · n8n and AI agents

Explain what n8n and AI agents are, and build workflows using triggers and actions.

2

LO2 · Webhook-triggered agents

Implement AI agent applications triggered by webhooks and integrated with business systems.

3

LO3 · Agentic chatbots

Design agentic chatbot workflows that operate within business and regulatory constraints.

4

LO4 · Human in the loop

Apply human approval so that sensitive actions are authorised by a named person.

5

LO5 · RAG

Deploy Retrieval-Augmented Generation to ground an agent in your organisation's documents.

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COURSE JOURNEY

Five Connected Labs

Each lab keeps the agent and changes one thing — the front door, the tools, the oversight, the knowledge.

LAB 1
form → email

▶

LAB 2
webhook + agent + tools

▶

LAB 3
chatbot + memory

▶

LAB 4
human approval gate

▶

LAB 5
RAG over documents

1 The front door changes

Form Trigger in Lab 1 becomes a Webhook in Labs 2–5 — the workflow behind it barely moves.

2 The agent gains abilities

Lab 2 adds tools, Lab 3 adds memory, Lab 5 adds retrieval over your own documents.

3 The oversight grows

Lab 4 stops the workflow dead until a licensed person approves what the agent drafted.

By Lab 5 you have built the whole pattern: a front door, an agent, its tools, its guardrails and its knowledge.

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WSQ FUNDING

Criteria for Funding

1

75% attendance

A minimum attendance rate of 75%, based on the SSG Digital Attendance record.

2

Assessed Competent

Complete the assessment and be assessed as 'Competent'.

3

Both are required

Both criteria must be met for the course fee funding to be granted.

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ASSESSMENT BRIEFING

Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black or blue pen for hard-copy assessments.
5. No liquid paper or correction tape.
6. Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

1. **Written Assessment (SAQ)**
   1 hour. Short-answer questions covering the course knowledge.

2. **Practical Performance (PP)**
   1 hour. Hands-on n8n build tasks drawn from the five labs.

3. **Open book**
   Slides, Learner Guide and approved materials only.

4. **75% attendance**
   A minimum of 75% attendance is required to be eligible for assessment and funding.

5. **Appeals**
   An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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FOUNDATION

Why Agentic AI, and Why Now

From rules to agents · what agentic automation is worth · where it is used

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FROM RULES TO AGENTS

The Evolution of AI

1

Artificial Intelligence · 1950s

Turing's test and symbolic, rule-based systems that encode human expertise by hand.

2

Machine Learning · 1990s

Data-driven systems that learn patterns from examples instead of being programmed with rules.

3

Deep Learning · 2000s

Neural networks — DNNs and CNNs — trained on large datasets with GPU compute.

4

Generative AI · 2020s

The 2017 Transformer and its attention mechanism made models that generate fluent text and images.

5

Agentic AI · 2024

LLMs given goals, tools and memory: they plan, act, observe and adapt rather than answering once.

6

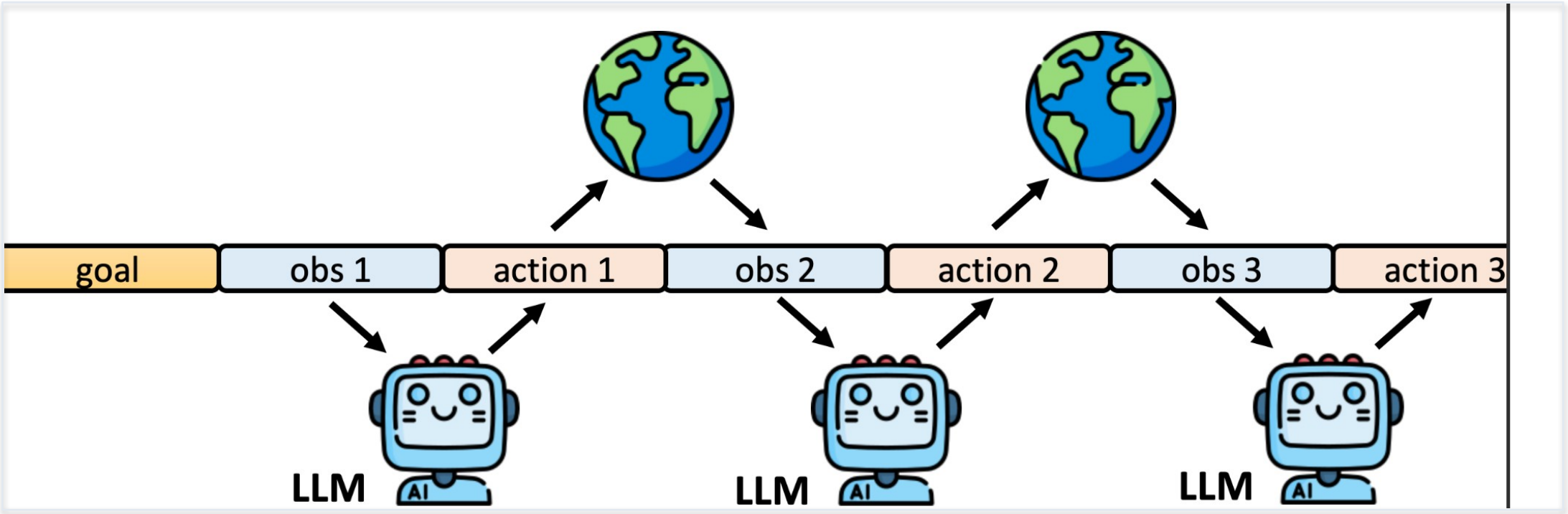
Physical AI · 2025

Agents that perceive and act in the physical world — robotics and embodied systems.

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THE AGENT LOOP

How an AI Agent Works

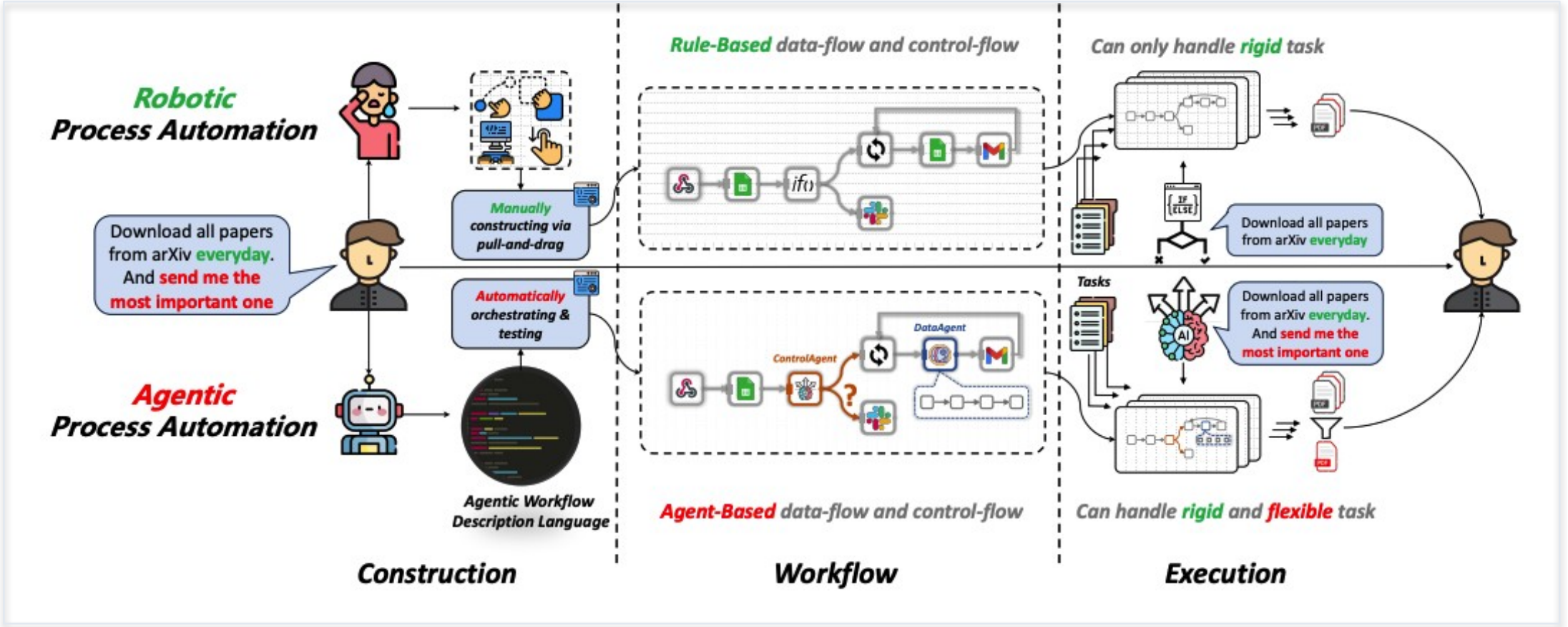


The agent loop: given a goal, the LLM observes, chooses an action, acts on the world through a tool, observes the result, and repeats until the goal is met.

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APA

From RPA to Agentic Process Automation



Robotic Process Automation follows a rigid path you construct by hand. Agentic Process Automation lets an agent orchestrate the workflow, so it handles both rigid and flexible tasks.

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WHY ORGANISATIONS ADOPT IT

Business Value of Agentic Automation

Six reasons organisations move from hand-built automation to agents.

1 Free people from repetitive admin

Scheduling, form processing, data entry, report generation and email routing run themselves.

2 Shift effort to high-value work

When the admin burden lifts, people spend their time on judgement, relationships and strategy.

3 Respond in real time

Events — a form, a payment, a customer message — trigger a workflow the moment they happen.

4 Work around the clock

An agent answers a 2am enquiry with the same care as a 2pm one, and never forgets to log it.

5 Reduce human error

The same rules are applied to every case, and every decision is written to an audit trail.

6 Scale without headcount

Handling ten times the volume costs compute, not another ten hires.

The gain is not speed alone — it is consistency, traceability and capacity you did not have before.

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USE CASES

Where Agentic Automation Is Used

The same pattern — trigger, agent, tools, oversight — recurs across every function.

1 Customer service

Answer enquiries from your own documents, capture leads, and escalate what needs a person.

2 Finance & accounting

Invoice processing, bank reconciliation, accounts-payable routing and exception handling.

3 Risk & compliance

Screen applications, flag anomalies, and route sensitive cases for human review.

4 HR & people ops

Onboarding, leave applications, policy questions and approval chains.

5 Sales & marketing

Lead qualification, follow-up sequencing, CRM enrichment and content drafting.

6 Operations

Order processing, status notifications, document generation and reporting.

Every one of these is a front door, an agent, some tools and a decision about where the human belongs.

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COMPLEXITY VS CUSTOMISATION

Where AI Agent Platforms Sit

Three tiers, and why this course teaches the middle one.

1 No-code

Fully visual builders. Fastest route to a working agent, least control over the internals.

2 Low-code · n8n

A visual canvas plus code nodes, 400+ integrations, self-host or cloud. This course's platform.

3 Full-code

SDKs and frameworks such as LangChain. Maximum control, and real engineering effort.

n8n sits where a business analyst can build it and an engineer can still extend it.

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01

TOPIC 01

Overview of n8n and AI Agents

Workflow automation · Triggers & actions · LLM agents · Webhooks

Applied in Lab 1, Lab 2

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What Is n8n?

A workflow automation platform: connect apps, data and AI on one visual canvas — no-code to full-code.

1 Build visually

Drag nodes onto a canvas and wire them together. 400+ app integrations, plus HTTP and Code nodes when no integration exists.

2 Run anywhere

Free cloud trial at n8n.io, or self-host with Docker. Webhooks, schedules and chat triggers work the same in both.

3 AI-native

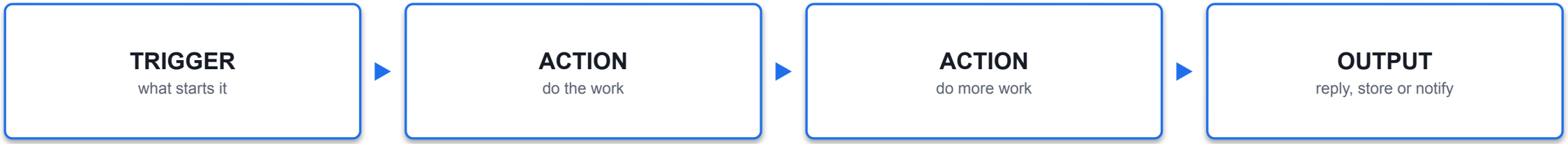
AI Agent and LangChain nodes are built in — attach any LLM (Gemini, OpenAI, Claude), memory, tools and vector stores.

One canvas to connect apps, data and AI — automation and agents without gluing code by hand.

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Every Workflow Is the Same Shape

Trigger first, then actions in order, with data flowing left to right. Only the nodes change.



1 One trigger

A workflow has exactly one entry point. Change the trigger and you have a different workflow — the actions may not change at all.

2 Actions are ordered

Each action runs after the one before it, and can read the output of every node that ran before it.

3 Data flows as JSON

Every node passes a list of JSON items to the next — the same structure in every lab of this course.

Trigger, actions, output. Learn this once in Lab 1 and every later lab is a variation.

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Trigger Nodes — What Is Allowed to Start a Workflow

The trigger you choose is a statement about who or what may start the process.

1 Manual Trigger

Click Execute workflow. For building and testing, and for one-off jobs such as the Lab 5 ingestion run.

2 Schedule Trigger

A clock starts the workflow — an interval or a cron expression. Nightly reports, reminders and syncs.

3 Form Trigger

n8n hosts a web form for you and each submission starts a run. This is Lab 1's front door.

4 Webhook

An external system calls a URL you expose. Your own website, app or CRM — Labs 2, 3, 4 and 5.

5 Chat Trigger

Each message in n8n's built-in chat window starts a run — the quickest way to talk to an agent while building.

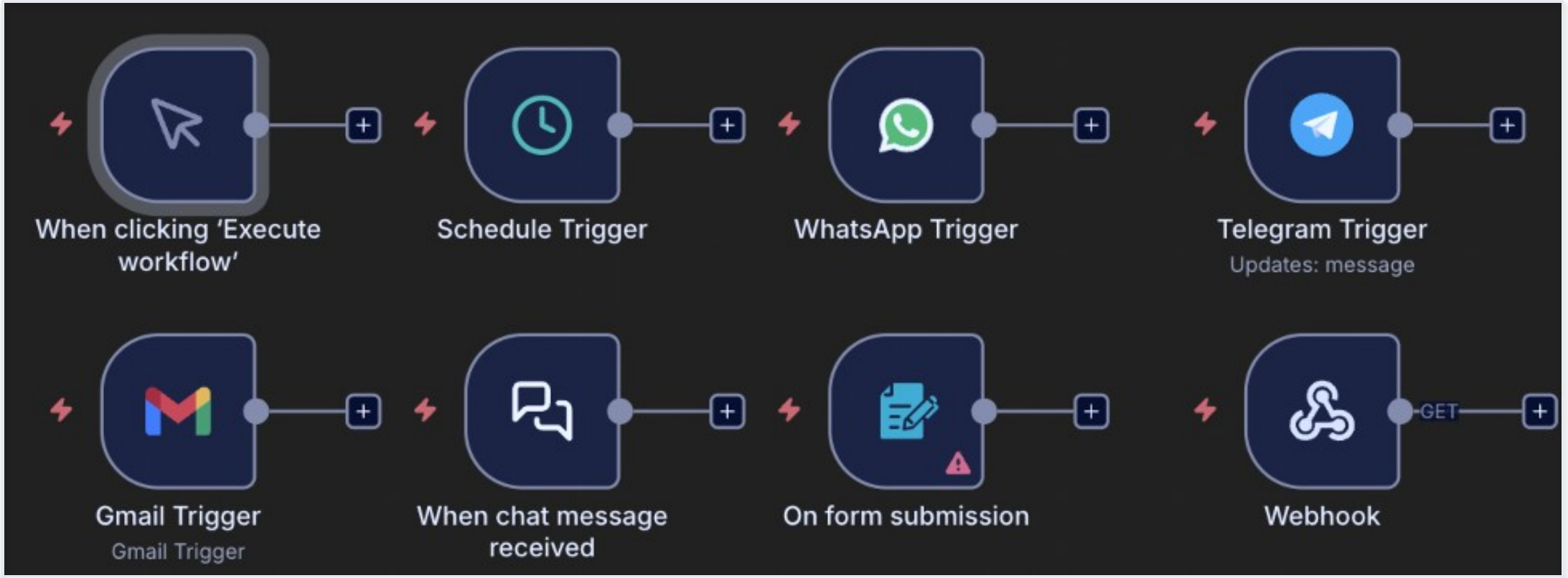
6 App & Email Triggers

On new Gmail, new Sheets row, new Drive file. React to changes in the apps you already use.

Pick the trigger by asking: who or what should be allowed to start this process?

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Trigger Nodes on the n8n Canvas



n8n trigger nodes: Manual · Schedule · WhatsApp · Telegram · Gmail · Chat · Form · Webhook

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Triggers Used in This Course

Every lab's trigger, and why that trigger is the right front door for that business process.

Trigger	Where	Why it fits
Form Trigger	Lab 1	A visitor's event sign-up must start the workflow
Webhook — POST	Labs 2, 3, 4, 5	A website sends JSON and waits for the agent's reply
Manual Trigger	Lab 5 ingestion	A human loads the documents once, by hand
Schedule Trigger	Not used in class	Nightly reports and reminders — same pattern, a clock instead
Chat Trigger	Optional while building	Talk to an agent without building a website first

Swap the front door and keep the workflow — the trigger is the only part that has to change.

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Data-Processing Nodes — The Workhorses

Between the trigger and the output, these nodes shape, route and transform the data.

1 Edit Fields (Set)

Add, rename or reshape fields — the node you reach for first when the data is the wrong shape. Used in Labs 2 and 4.

2 IF / Switch

Branch the workflow. IF gives a true and a false path; Switch gives one path per case. Lab 4 routes Approve vs Decline.

3 Filter

Let only the matching items continue and drop the rest of the list.

4 Merge

Combine two branches — append the rows, match them by key, or choose one side.

5 Split Out / Aggregate

Turn one item holding a list into many items — and back again.

6 Code

JavaScript or Python for the 5% a built-in node cannot do. Lab 3 uses it to normalise the chat message.

7 Loop / Wait

Process items in batches, or pause a workflow for a period of time or an event.

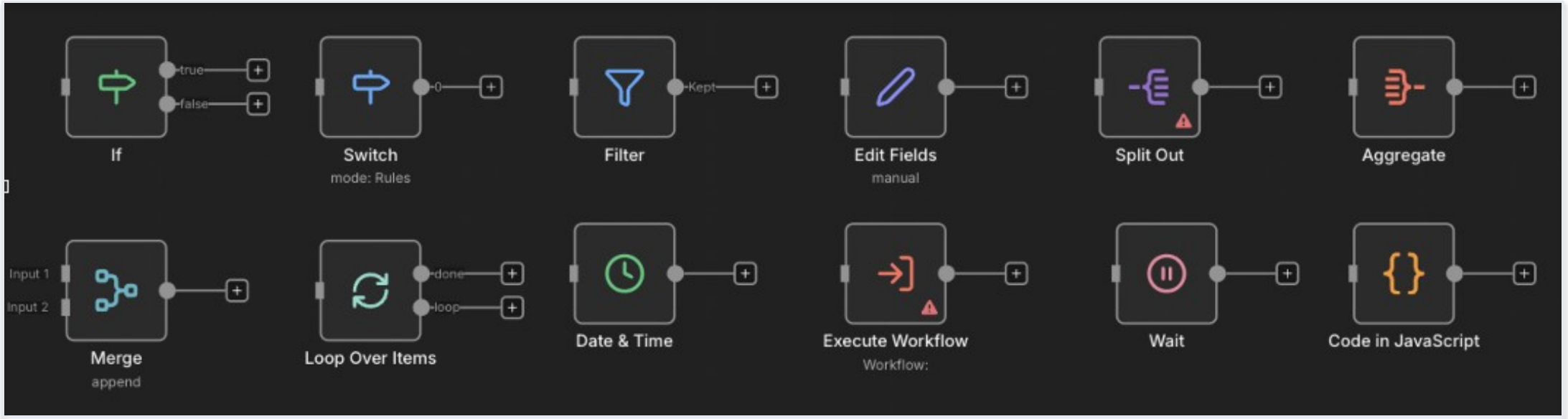
8 HTTP Request

Call any API on the internet — the universal escape hatch when no integration exists.

Most workflows are: trigger → reshape (Set) → decide (IF) → act (app node). Reach for Code last.

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Data-Processing Nodes on the Canvas



If · Switch · Filter · Edit Fields · Split Out · Aggregate · Merge · Loop · Date & Time · Wait · Code

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Where the Data Goes — Storage & App Actions

Action nodes that write the result somewhere useful, or send it to a person.

1 Google Sheets

A real spreadsheet you keep, connected with OAuth2. Append Row is the workhorse — Labs 2 and 4 log every decision here.

2 Gmail / Outlook

Send a message — and in Lab 4, Send and Wait for Response, which turns email into an approval gate.

3 Vector Store

Store embeddings for RAG. In-memory for class, Pinecone / Supabase / Qdrant for production — Lab 5.

4 Respond to Webhook

Return data to whoever called the webhook, so a web page can render the agent's answer.

Anything you must keep goes to external storage — an in-memory store vanishes when n8n restarts.

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JSON and Expressions — How Data Moves

Every node passes a list of items; each item is a JSON object of key/value pairs.

- JSON (JavaScript Object Notation) is a simple text format for data.
- Values can be text, numbers, true/false, lists [ ] or nested objects { }.
- Most APIs send and receive JSON — and so does every n8n node.
- Click any node to inspect the JSON going in and coming out.

ONE FORM SUBMISSION AS JSON

```
{
  "Name": "Alice Tan",
  "Email": "alice@corp.com",
  "Phone": "91234567",
  "Message": "Please reserve 2 seats"
}
```

Read any value later with an expression: {{ $json.Name }} → "Alice Tan"

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Test vs Production Executions

The same workflow runs in two modes — always know which one you are looking at.

TEST — in the editor

- Click Execute workflow to run it once.
- The data is shown live on the canvas, node by node.
- Pin a node's output to freeze it while you build downstream.
- Perfect while building — nothing needs to be Active.

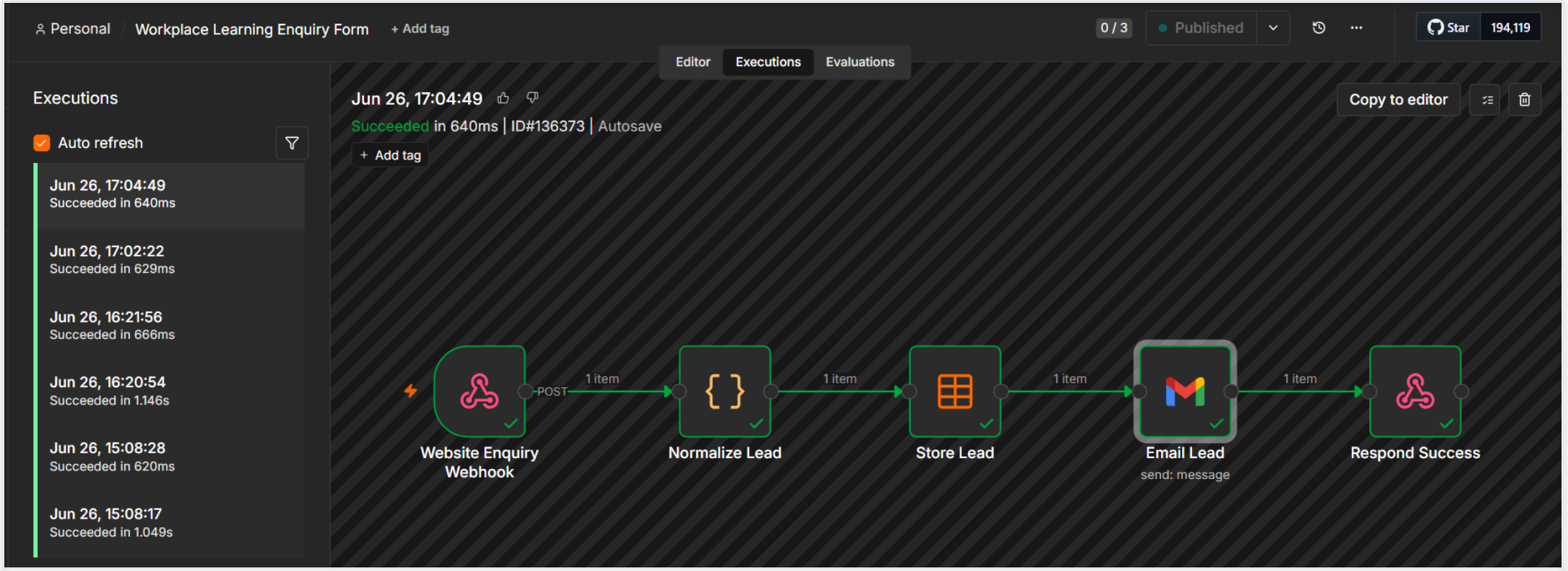
PRODUCTION — Active toggle ON

- Runs automatically whenever the trigger fires.
- Nothing appears on the canvas — runs are logged under Executions.
- Every run records status, timing and the full data at each node.
- This is what your users actually experience.

The Executions list is the honest account of what happened — debug from there, not from memory.

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Execution History



The Executions tab — every run shows its status, duration, ID and the full data at each node

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Credentials — Connect Your Accounts Once

Nodes reference credentials by NAME. Set each one up once under Credentials → Add credential.

1 Gmail (OAuth2)

Sign in with Google and grant access. Used from Lab 1 onward for sending mail and for the Lab 4 approval gate.

2 Google Gemini (API key)

The LLM behind every agent in this course, and the embedding model in Lab 5.

3 Google Sheets (OAuth2)

The audit log in Labs 2 and 4 — customer records, decisions, drafts and approvals.

4 Google Drive (OAuth2)

The brochure folder that Lab 5's ingestion workflow reads from.

Imported workflows carry credential NAMES, not secrets — always re-select your own after importing.

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Generative AI vs Agentic AI

The difference is not how clever the model is — it is whether it can act.

GENERATIVE AI — answers

- You ask; it writes one answer.
- No access to your systems, and no actions.
- Quality depends on what you paste into the prompt.
- Stops the moment the answer is produced.
- Example: draft me an email to this customer.

AGENTIC AI — acts

- Works toward a goal, not just one reply.
- Calls tools: looks up data, calls APIs, writes records.
- Holds a conversation with memory.
- Continues until the goal is met.
- Example: read the enquiry, check the CRM, reply and log it.

Generative AI produces content. Agentic AI decides and does — which is exactly why it needs governance.

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Components of an AI Agent

Four parts around one hub: the AI Agent node wires them together, and every agent in this course is exactly these four.

MODEL

The LLM 'brain' — interprets language, reasons and chooses the next step. Google Gemini in this course.

MEMORY

Keeps the conversation so follow-up questions still make sense. Keyed on a session ID, per visitor.

THE AI AGENT

plans · decides · calls tools

TOOLS

The abilities it may use: look up a Sheets row, search a vector store, send an email, call an API.

INSTRUCTIONS

The system message — identity, scope, rules and refusals. Always in force, on every single turn.

Model + Memory + Tools + Instructions. Change any one of the four and you have a different agent.

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The Model — Popular LLM Providers

Any chat model plugs into the same AI Agent node — swap the brain and keep the agent.

Provider	Models	Notes
Google	Gemini 2.5 Flash / Pro	Course default — also supplies the embeddings in Lab 5
OpenAI	gpt-4.1, gpt-4.1-mini, gpt-4o	Widely used; mini tier is cheap and fast
Anthropic	Claude (Opus, Sonnet, Haiku)	Strong reasoning and long context
Meta	Llama 3.x	Open weights — can be self-hosted
Mistral	Mistral / Mixtral	Open weights — a European option

Pick by task: small fast models for classification and guardrails, larger models for reasoning.

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Model Parameters — Temperature and Friends

Settings on the chat-model node shape HOW the model answers. The most important by far is temperature.

1 Temperature 0 – 0.3

Deterministic and repeatable. Use for classification, compliance decisions and data extraction — automation that must give the same answer twice.

2 Temperature ≈ 0.7

Balanced. Natural-sounding conversation with some variety — the usual default for a customer-facing assistant.

3 Temperature 1.0 +

Creative and surprising. Marketing copy and brainstorming — never for a decision that must be defended.

1 Top-p

An alternative sampling control that limits the model to the most probable tokens. Tune temperature OR top-p, never both.

2 Max tokens

Caps the length of the reply. Protects your cost and keeps chat answers short enough to read.

3 Frequency / presence penalty

Discourage repetition and nudge the model onto new topics — useful for long generations, rarely for automation.

For business automation keep the temperature LOW — Lab 3 uses 0.2 so the same question is refused the same way every time.

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Memory — Why the Agent Remembers

Without memory every message is from a stranger; with it, 'what about next month?' still makes sense.

Msg 1: 'What courses do you run?'
agent answers from its tools

▶

Stored in memory
keyed by session ID

▶

Msg 2: 'How much is that one?'
agent knows which course

▶

Window slides
only the last N turns are kept

1 Window buffer memory

Keeps the last N exchanges of THIS conversation. It is n8n's default and it is what Lab 3's chatbot uses.

2 Session key

Memory is per session ID, so two visitors on the same website never see each other's conversation.

3 Sometimes you want NO memory

Lab 2's onboarding agent deliberately has none — one applicant's details must never influence the next decision.

Memory is conversation context, NOT long-term knowledge. Facts belong in tools and documents (RAG).

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Tools — What the Agent May Do

Attach tools to the agent's Tool input. The model decides WHEN to call one, based on the description you write.

1 Google Sheets tool

Look up or write structured records. Lab 2 attaches three: check for a duplicate, create the customer, send the confirmation.

2 Vector Store tool

Retrieve document chunks for a grounded answer — the Query Data Tool in Lab 5.

3 HTTP Request tool

Call any external API in the middle of a conversation — prices, stock levels, a CRM record.

4 Sub-workflow tool

Wrap an entire n8n workflow as one callable tool, so your own logic becomes something the agent can invoke.

The tool DESCRIPTION is how the agent knows when to call it — write it for the model, not for a developer.

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Instructions — The Four Parts That Constrain

The system message is always in force. Every good agent instruction in this course states four things.

1 Identity

Who the agent is, and for whom. 'You are the onboarding assistant for Marina Trust Bank.' Narrow roles refuse far more reliably than general assistants.

2 Source rules

Where answers may come from. 'Answer ONLY from the brochures returned by the tool — never from general knowledge.'

3 Refusals

What is out of bounds, stated explicitly. 'Never recommend a product, predict a return, or guarantee an outcome.'

4 Escalation

What to do at the boundary. 'If you are not sure, say so and offer a consultation with a licensed adviser.'

Vague instructions produce confident wrong answers — every rule you skip, the model improvises.

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Examples of Good and Bad Instructions

The same chatbot, two system messages — one is testable, the other is a hope.

✗ TOO VAGUE

"You are a helpful assistant. Answer the user's questions politely."

No identity · no source rules · no refusals · no escalation.

It will happily invent a course fee and guarantee an investment return — politely.

✓ CONSTRAINED AND TESTABLE

"You are the enquiry assistant for Meridian Asset Management.

Answer only questions about our services, fees and appointment booking.

Use the brochure search tool for any factual question — never answer from general knowledge.

Never recommend a product, predict a return, or guarantee an outcome.

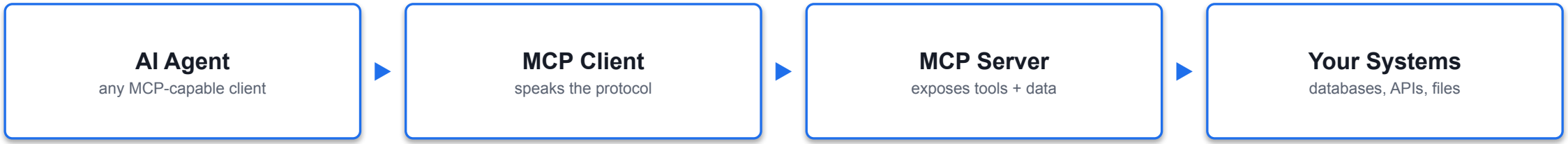
If you are not sure, say so and offer a call with a licensed adviser."

Identity · source rules · refusals · escalation — you can test each line by trying to break it.

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MCP — Model Context Protocol

An open standard that gives agents a common plug for tools and data, instead of one custom integration per pair.



Without MCP

Every agent × every tool needs its own custom integration. N agents and M tools means N × M connectors to build and to maintain.

With MCP

Build ONE MCP server per system and any MCP-capable agent can use it. N + M pieces, and the tools become reusable across agents.

MCP is like USB-C for AI tools — one connector, many devices.

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MCP in n8n

n8n sits on both sides of the protocol — it can serve tools to other agents, and borrow tools for its own.

MCP SERVER TRIGGER — n8n serves tools

- Expose your own n8n workflows AS tools over MCP.
- Claude, an IDE or another agent can then call your workflows directly.
- n8n becomes the tool layer for any MCP client.
- Secured with a bearer token on the server URL.

MCP CLIENT TOOL — n8n consumes tools

- Attach an external MCP server to your n8n AI Agent.
- The agent gains all of that server's tools with no custom node.
- Mix MCP tools and native n8n tools freely on the same agent.
- Same agent loop — simply more reach.

Two directions, one protocol. Build the tool once and every agent you own can use it.

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What Is a Webhook?

A URL your workflow listens on. When the outside world calls it, your workflow runs — the reverse of you calling an API.

Event happens
a visitor submits the form

▶

HTTP request to your URL
the website POSTs JSON

▶

Workflow runs
n8n receives the data as its input

▶

Respond to Webhook
your reply returns to the caller

1 Push, not pull

No polling and no schedule — the caller notifies you the moment something happens.

2 Any caller

Websites, payment providers, forms, CRMs, mobile apps — anything that can send an HTTP request.

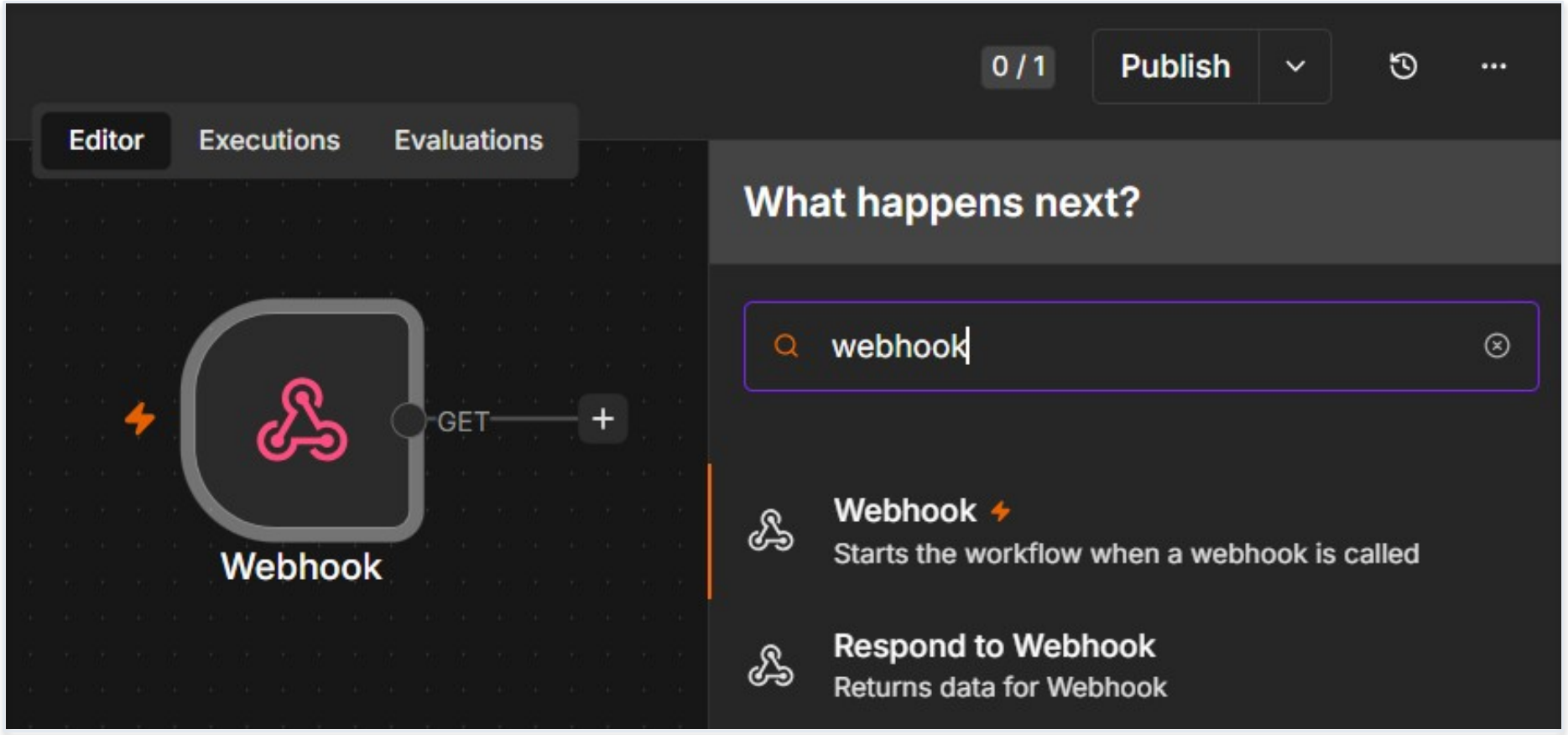
3 A paired node

The Webhook node starts the workflow; Respond to Webhook sends the answer back on the same connection.

A webhook is the reverse of an API call — the outside world calls YOU when an event happens.

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Webhook Nodes in n8n



The Webhook node starts the workflow when it is called; Respond to Webhook returns data to the caller

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Webhook vs HTTP Request — Which Way the Call Goes

Both are plain HTTP. The only difference is who dials.

WEBHOOK — inbound

- An external service calls YOUR workflow.
- You react the moment the event happens.
- It is the TRIGGER — always the first node.
- Labs 2–5: the lab website calls your agent.

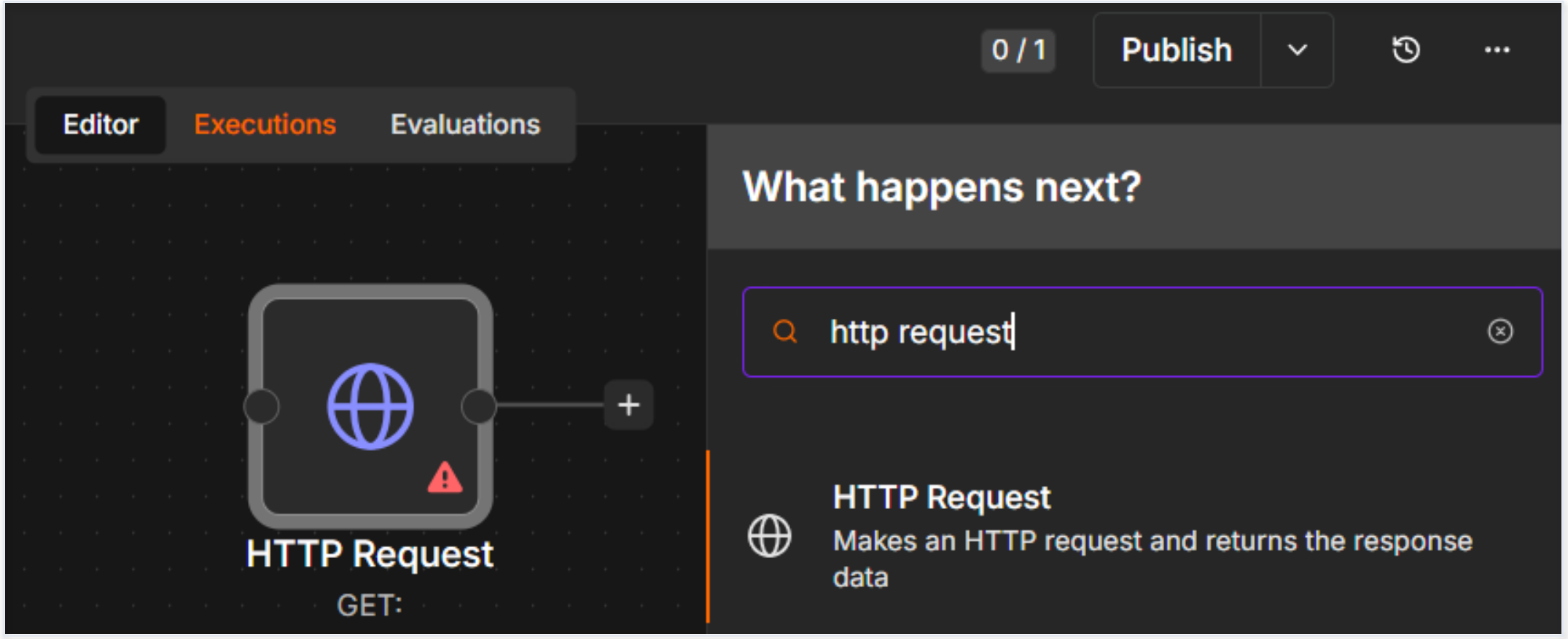
HTTP REQUEST — outbound

- YOUR workflow calls an external service.
- You pull the data at a moment you choose.
- It is an ACTION inside the workflow.
- Use it to reach any API with no dedicated n8n node.

Same HTTP underneath — a webhook receives a call, an HTTP Request makes one.

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The HTTP Request Node



The HTTP Request node calls out to any API — the outbound counterpart to an inbound webhook

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HTTP Methods on a Webhook — GET vs POST

The method is part of the contract with the caller: where the data rides, and what the call means.

GET — read

- The data rides in the URL query string:
  - ?email=alice@corp.com
- A browser can call it just by visiting the URL.
- Read it in n8n as {{ $json.query.email }}.
- Right for: a dashboard pulling data to display.

POST — send

- The data rides in the request body as JSON:
  - { "message": "How much is the course?" }
- This is what a web form or a chat widget sends.
- Read it in n8n as {{ $json.body.message }}.
- Right for: Labs 2–5, where the site submits data.

GET reads, POST sends. Set the method on the Webhook node — and the caller must match it exactly.

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Webhook Test URL vs Production URL

Every Webhook node gives you TWO URLs. They behave differently, and confusing them is the single most common webhook bug.

TEST URL — /webhook-test/…

- Works only while you have clicked 'Listen for test event'.
- Handles exactly ONE call, then stops listening.
- The run appears live on the canvas — ideal for debugging.
- Dies the moment you leave the editor.

PRODUCTION URL — /webhook/…

- Requires the workflow to be toggled Active.
- Always on — every call starts a run.
- Runs are logged under Executions, not drawn on the canvas.
- This is the URL you give to a website, an app or a QR code.

Your website must call the PRODUCTION URL of an ACTIVE workflow — the Test URL is spent after one request.

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Configuring the Webhook Node

Four settings decide whether the browser's call actually succeeds.

1 Method + Path

GET or POST, and the path that ends your URL — for example POST /webhook/marina-trust-onboarding.

2 Allowed Origins (CORS)

Browsers block cross-site calls unless you allow them. Set * in class so a local HTML page can call n8n; never leave * in production.

3 Response mode

Immediately (a bare acknowledgement), when the workflow finishes, or via a Respond to Webhook node — every lab here uses the last one.

4 Authentication

None for a classroom demo; Basic, Header or JWT for anything real. A public URL is a public door, and every call costs you model tokens.

'Failed to fetch' in the browser is almost always CORS or a dead Test URL — check those two first.

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Structured Output — Making an Agent's Answer Usable

By default an agent returns a paragraph of prose. Prose cannot be branched on, sorted, counted or coloured.

1 Attach an Output Parser

A Structured Output Parser forces named fields instead of free text — Labs 2 and 4 both use one.

2 Name the fields you will act on

Lab 2 returns decision, reason and riskFlags. The website colours its result card from decision alone.

3 Enumerate the allowed values

APPROVED / REJECTED / REVIEW / DUPLICATE. A closed list is what makes an IF node reliable.

4 Everything downstream depends on it

An IF node routes on it, a Sheets node logs it, and a queue is sorted by it. Prose supports none of that.

If something downstream must act on the answer, make the answer a field — not a sentence.

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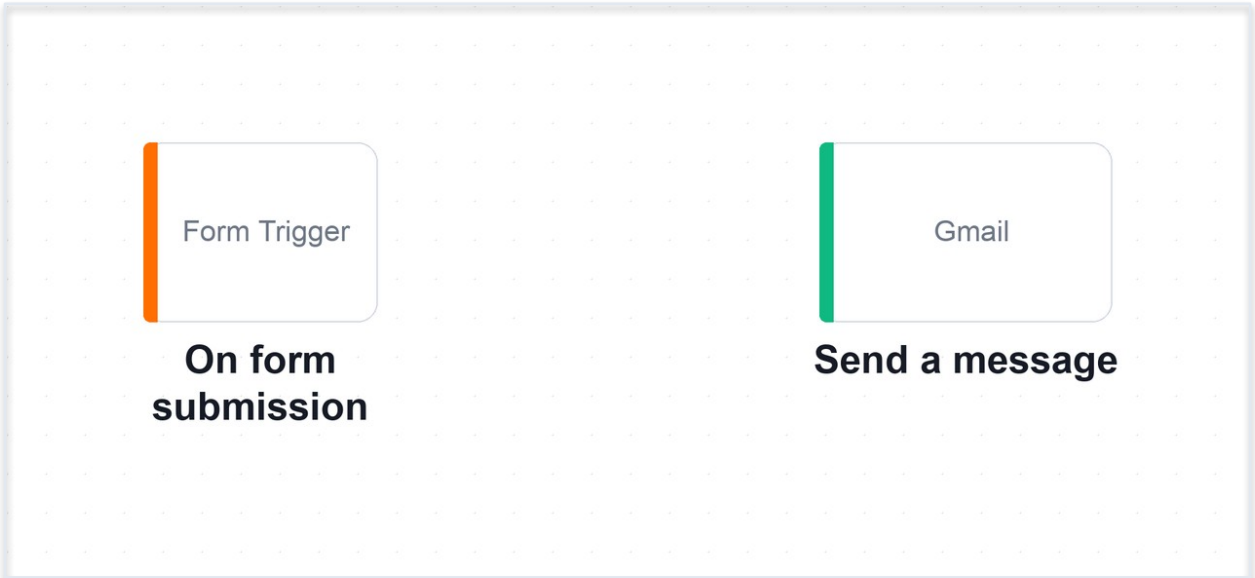
Lab 1 — n8n Trigger and Action (Event Flyer with QR Code)

Build your first n8n workflow end to end: a Form Trigger collects a visitor's details and a Gmail action emails them to the event administrator.

Form Trigger
visitor submits the sign-up

▶

Gmail — Send a message
administrator is emailed



The Lab 1 workflow on the n8n canvas. Two nodes: the Form Trigger decides WHEN the workflow runs, and the Gmail node decides WHAT it does.

LAB 1

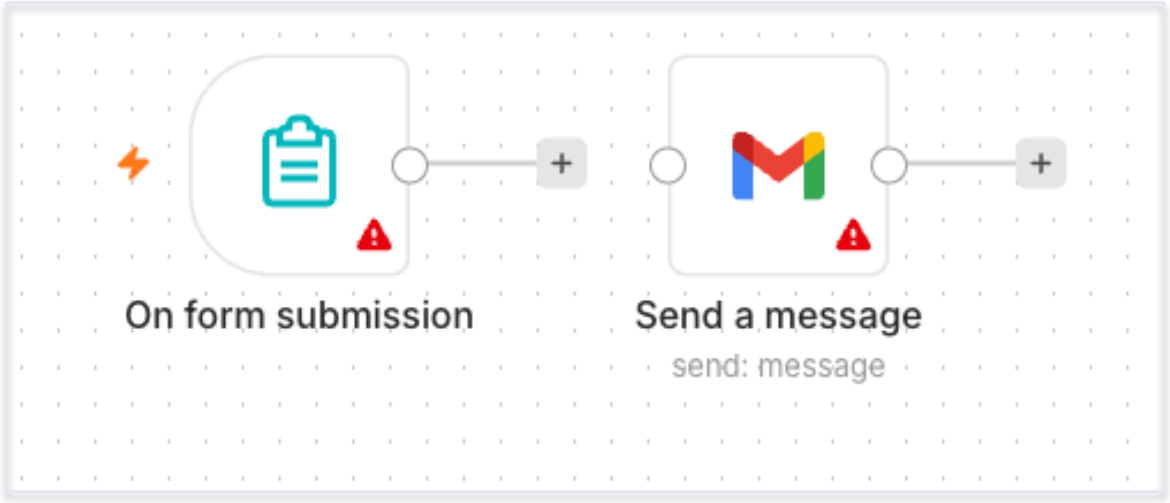
n8n Trigger and Action (Event Flyer with QR Code)

Form Trigger → Gmail (Send a message) · full click-by-click steps in the Learner Guide

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Lab 1 — n8n Trigger and Action — What You'll Build

- The finished Lab 1 workflow on the n8n canvas — the Form Trigger fires on each submission and the Gmail node sends the details to the administrator.
- A trigger decides when a workflow runs; an action decides what it does.
- Expressions such as {{ $json.Name }} carry data from one node to the next.



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Lab 1 — n8n Trigger and Action — What You'll Build

- A sample event flyer from a previous class. The QR code encodes the form's Production URL, so anyone can sign up by scanning it with a phone.



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LAB 1 · VERIFY AND DEBRIEF

Lab 1 — n8n Trigger and Action

Scan the QR code on your flyer with your phone, submit the form, and confirm the administrator's inbox receives the email with the details you typed. A reference solution is provided as labs/lab1-trigger-actions/lab-trigger-actions.json — build yours by hand first, then import it to compare.

1 Triggers and actions

A trigger decides when a workflow runs; an action decides what it does.

2 Expressions

Expressions such as {{ $json.Name }} carry data from one node to the next.

3 Test vs Production URL

The Test URL works only while you are listening; the Production URL works once the workflow is Active.

Lab 1 — you can now build an automated workflow in n8n using a trigger and an action.

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Lab 2 — Webhook and AI Agent (Marina Trust Bank Onboarding)

Replace the n8n form with a real bank website that POSTs to a webhook, where an AI Agent applies the bank's onboarding rules using three tools.

Webhook — POST
the bank site submits

▶

Normalise
Set node tidies the fields

▶

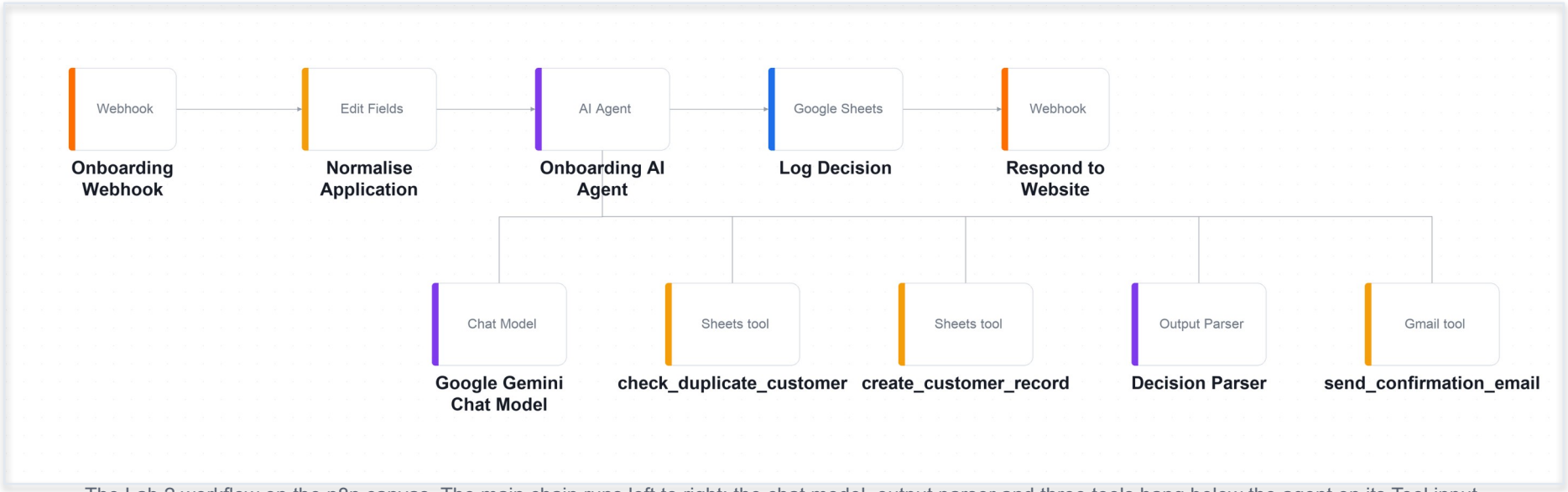
AI Agent
+ 3 tools + parser

▶

Log Decision
Google Sheets audit row

▶

Respond
decision returns to the page



The Lab 2 workflow on the n8n canvas. The main chain runs left to right; the chat model, output parser and three tools hang below the agent on its Tool input.

LAB 2

Webhook and AI Agent (Marina Trust Bank Onboarding)

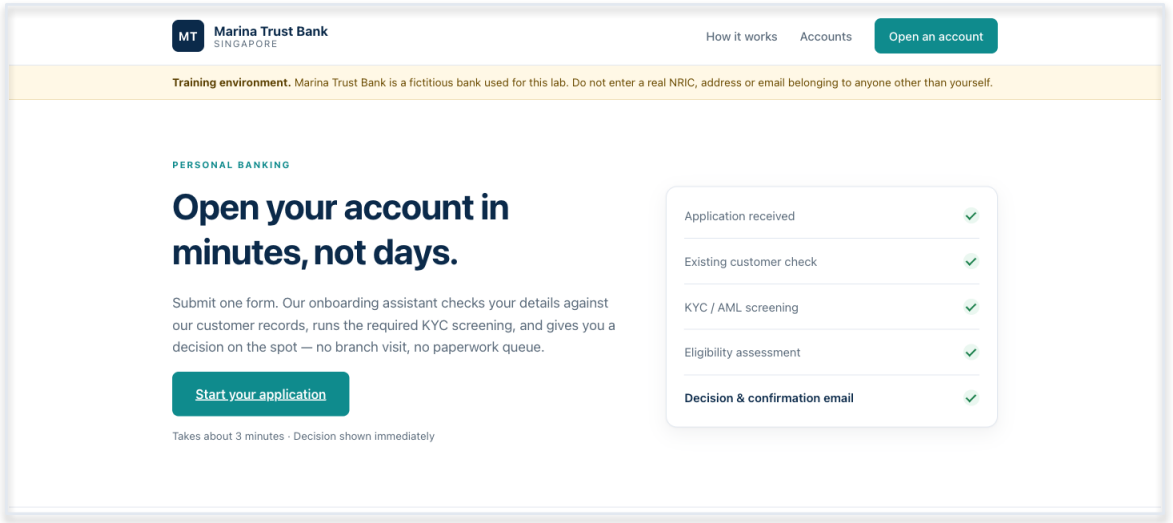
Webhook → Normalise → AI Agent (+3 tools) → Log Decision → Respond to Website · full click-by-click steps in the Learner Guide

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LAB 2 · THE APPLICATION

Lab 2 — Webhook and AI Agent — What You'll Build

- The Marina Trust Bank site the webhook sits behind. The applicant submits here, the AI Agent decides, and the page renders the decision as a coloured card.
- A webhook lets any external system — your website, an app, a CRM — start an agent workflow.
- Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.



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LAB 2 · VERIFY AND DEBRIEF

Lab 2 — Webhook and AI Agent

Run TC1 (a new applicant) and confirm a green APPROVED card, a new row in the Customers sheet and a confirmation email. Then run TC2 (an already-seeded NRIC) and confirm an amber DUPLICATE card with no new row.

1 Webhooks

A webhook lets any external system — your website, an app, a CRM — start an agent workflow.

2 Tools make an agent

Tools are what turn an LLM into an agent: it decides when to look up a customer and when to create one.

3 Structured output

Structured output (decision, reason, riskFlags) is what lets downstream nodes branch, log and colour a result.

4 The front door is separate

The agent is unchanged from a form-triggered version — only the front door differs.

Lab 2 — you can now deploy an AI agent that is triggered by a webhook from an external website.

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Lunch Break

1 hour

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02

TOPIC 02

Agentic Chatbot and Human in the Loop

Agentic workflows · Chatbots on webhooks · Approval gates · Accountability

Applied in Lab 3, Lab 4

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Human In, On, and Out of the Loop

Three levels of human oversight. The design question is which one this particular process deserves.

1 Human IN the loop

The workflow PAUSES and a named person must approve before the action happens. Nothing reaches the customer unauthorised. This is Lab 4.

2 Human ON the loop

The agent acts immediately, but a person monitors and can intervene or reverse it. Suited to high volume and low individual risk.

3 Human OUT of the loop

Fully autonomous. Acceptable only where an error is cheap, reversible and detectable — never where money or a regulator is involved.

The question is never 'is the model good enough?' but 'what does one bad output cost, and who answers for it?'

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How an Approval Suspends a Running Workflow

A Send-and-Wait node stops the execution mid-flight and holds it until a person decides.

Event arrives
a client message

▶

Agent drafts
classify, flag, propose

▶

Draft is logged
before any human sees it

▶

Human decides
Approve or Decline

▶

Only then
send and log the outcome

1 The execution really is paused

Look at the canvas and the run is sitting on the approval node. It waits days if it has to — n8n holds the state.

2 Approve and Decline are just links

The email carries two URLs. Clicking one resumes the execution down that branch — no app for the approver to learn.

3 A decline is a reassignment

Declining is not deleting. The ticket moves to a named person, because someone still owes the client an answer.

The agent may do everything except the one thing that matters. That pause is the whole control.

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When to Put a Human in the Loop

Six tests. If a process meets any one of them, it needs an approval gate.

1 Money moves

Payments, refunds, credit and lending decisions — anything with a direct financial consequence.

2 It is a regulated communication

A licensed firm writing to a client about their money, health or legal position.

3 The action is irreversible

A sent email, a deleted record or published content cannot be recalled.

4 A person is vulnerable

Distress, illness, bereavement, or savings that someone cannot afford to lose.

5 The stakes exceed the error rate

A 2% error rate is fine for tagging tickets and unacceptable for approving loans.

6 Someone must be accountable

A regulator asks which licensed person is answerable — not what the machine decided.

Automate the drafting, not the deciding, wherever any of these six is true.

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Guardrails — Two Gates Around the Agent

Instructions shape what the agent intends. Guardrails check what actually goes in and comes out.

INPUT GUARDRAIL — before the agent

- Screen the incoming message before the model ever sees it.
- Block prompt injection: 'ignore your instructions and…'.
- Strip or reject personal data you must not process.
- Rate-limit and authenticate — a public webhook is a public door.

OUTPUT GUARDRAIL — after the agent

- Check the reply before it reaches a human being.
- Look for the phrases you have banned: guarantees, forecasts, advice.
- Validate the structured fields against the values you allow.
- On any doubt, route to a person rather than sending it.

Defence in depth: the same rule in the prompt AND in the workflow — only one of the two runs on a machine you control.

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Lab 3 — Chatbot Webhook (Investment Advisor)

A compliance-bound chatbot for an investment advisory firm: no answer until the firm can contact the visitor, and never any financial advice.

Chat Webhook — POST
the widget sends a message

▶

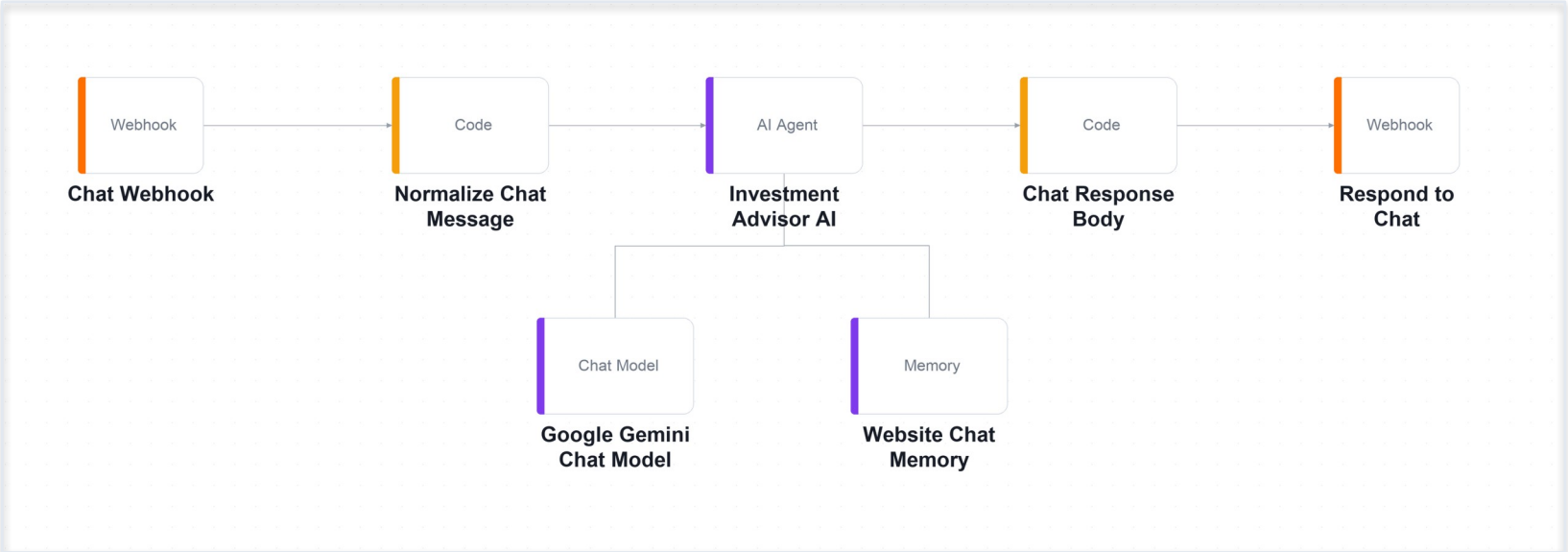
Normalize
Code node shapes the input

▶

AI Agent
Gemini 0.2 + window memory

▶

Respond to Chat
reply returns to the widget



The Lab 3 workflow on the n8n canvas. A small graph: webhook in, normalise, the agent with its chat model and session memory, then the reply back out.

LAB 3

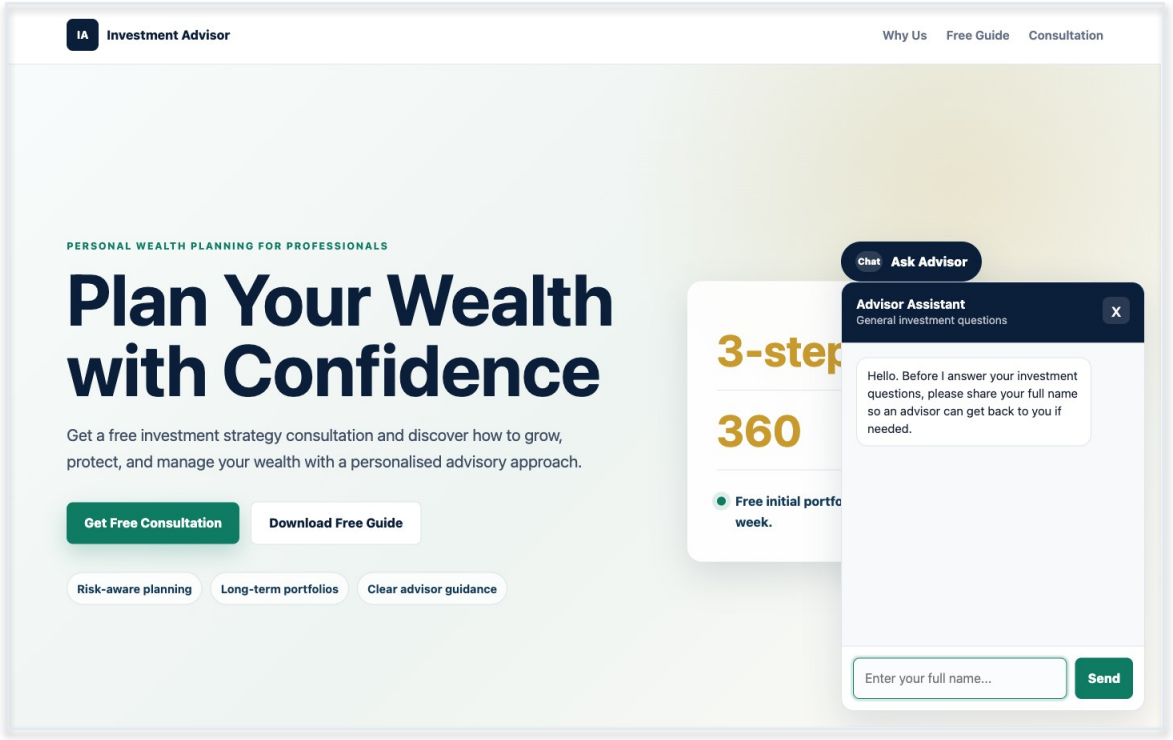
Chatbot Webhook (Investment Advisor)

Chat Webhook → Normalize → AI Agent (+ model + memory) → Respond to Chat · full click-by-click steps in the Learner Guide

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Lab 3 — Chatbot Webhook — What You'll Build

- The Investment Advisor site with the chat widget open. Note the contact gate in action — it asks for the visitor's name before it will answer anything.
- An agentic chatbot's system message is a compliance control, not just a prompt.
- The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.



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LAB 3 · VERIFY AND DEBRIEF

Lab 3 — Chatbot Webhook

Ask 'Can you guarantee returns?' and 'Which stock should I buy?'. The agent must decline both and offer a consultation. Then ask a question before giving your details — it must ask for the missing detail and answer nothing.

1 The system message

An agentic chatbot's system message is a compliance control, not just a prompt.

2 Defence in depth

The same rule enforced in the browser and in the agent is deliberate redundancy — only one of them runs on a machine you control.

3 Session-keyed memory

Session-keyed memory gives context to follow-up questions without leaking between visitors.

4 Prompt or RAG?

A small fixed FAQ belongs in the prompt; RAG is for knowledge too large or too changeable for that.

Lab 3 — you can now deploy an agentic chatbot on a website that operates within regulatory limits.

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Lab 4 — Human in the Loop (Client Rapport Assistant)

An agent reads a worried client's concern, flags compliance risks and drafts a reply — but sends nothing until a licensed manager approves it.

Webhook
client sends a concern

▶

AI Agent
classify · flag · draft

▶

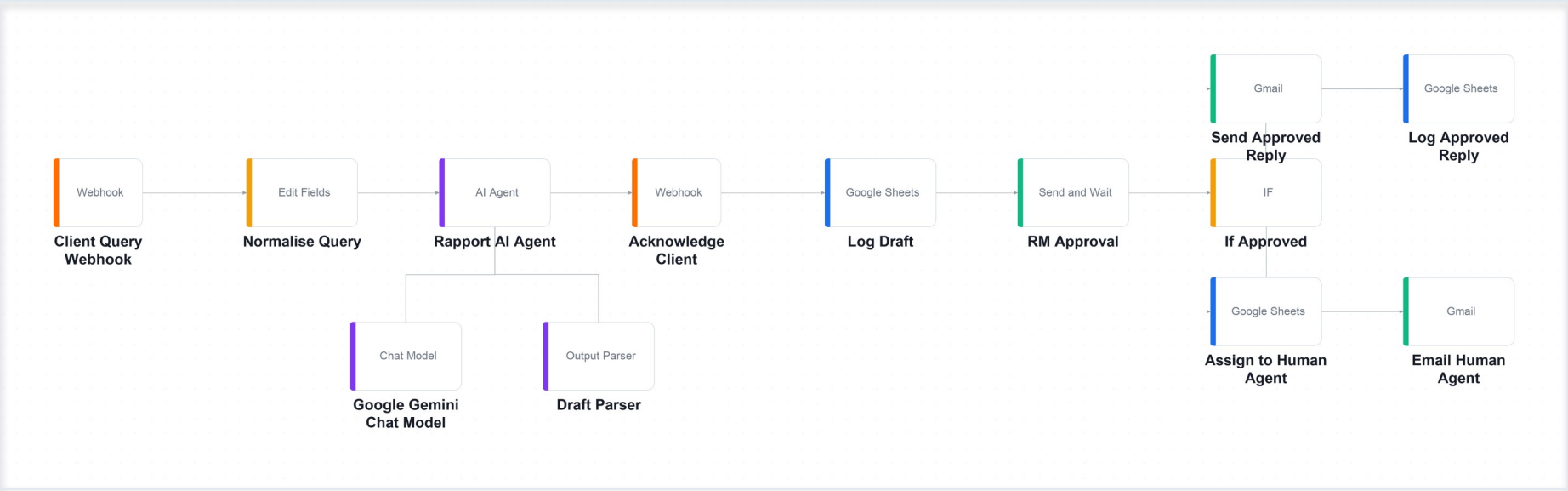
Log Draft
before any human sees it

▶

RM Approval
Send & Wait — PAUSED

▶

IF
Approve → send · Decline → reassign



Everything up to RM Approval is the agent's territory; the IF node after it splits into the approved path (top) and the reassignment path (bottom).

LAB 4

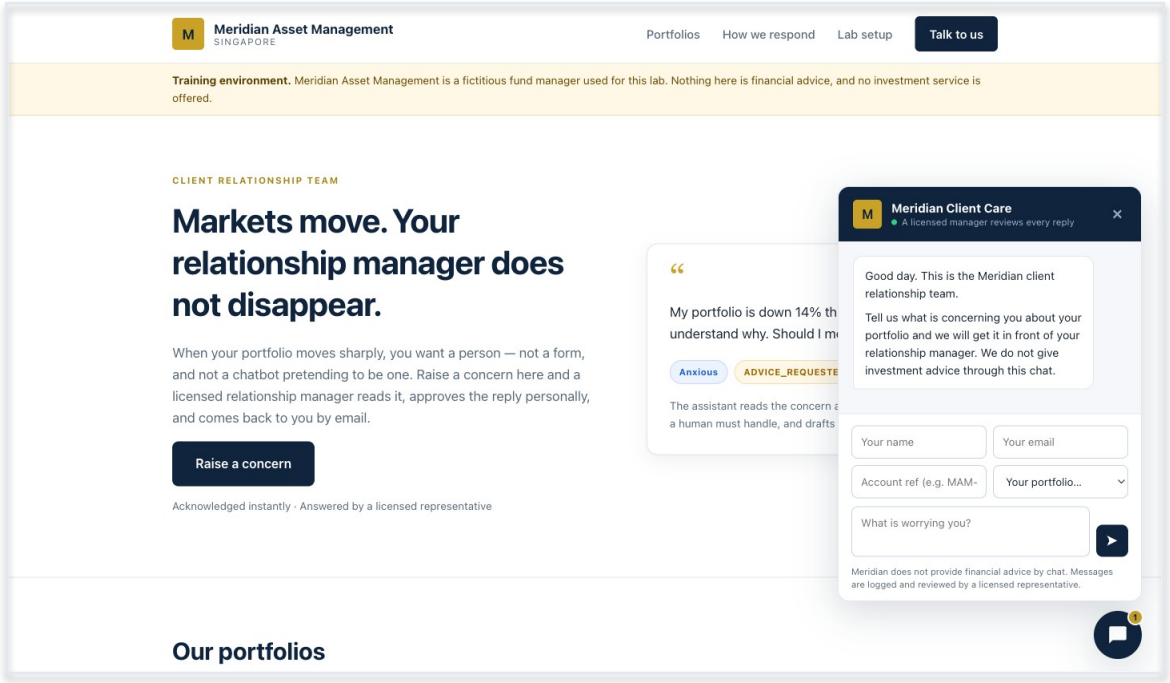
Human in the Loop (Client Rapport Assistant)

Webhook → Agent → Acknowledge → Log Draft → RM Approval (Send & Wait) → IF → Send / Reassign · full click-by-click steps in the Learner Guide

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Lab 4 — Human in the Loop — What You'll Build

- The Meridian client portal. The widget states plainly that a licensed manager reviews every reply — the human-in-the-loop promise made visible to the client.
- A Send-and-Wait node pauses the workflow until a named person authorises the action.
- Separating what the agent proposed from what a human approved is what makes the trail auditable.



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LAB 4 · VERIFY AND DEBRIEF

Lab 4 — Human in the Loop

Run all eight rows of sample-queries.csv. When you are done Drafts has 8 rows, Approved_Replies has 5 and Handover_Queue has 3 — and no accountability column is empty.

1 The approval gate

A Send-and-Wait node pauses the workflow until a named person authorises the action.

2 Two separate logs

Separating what the agent proposed from what a human approved is what makes the trail auditable.

3 A decline reassigns

A decline is a reassignment, not a deletion — someone must still owe the client an answer.

4 Keep the model out

The most legally important sentence in a workflow should be the one the model cannot touch.

Lab 4 — you can now apply a human approval gate so no AI-drafted communication is sent unsupervised.

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Tea Break

15 minutes

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03

TOPIC 03

Retrieval-Augmented Generation (RAG)

Tokenization · Embeddings · Vector stores · Grounded answers

Applied in Lab 5

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TOPIC 3 · RAG

Why Retrieval, Rather Than a Bigger Prompt

For twenty short documents, pasting them all would genuinely work. The problem starts at two hundred.

1 Prompts have a ceiling

Add enough documents and you exceed the model's context window — the request simply fails.

2 Long prompts degrade quality

Models attend less reliably to the middle of a very long context. The answer is in there and is still missed.

3 You pay for every token

Every question re-sends every document, whether or not it was relevant to that question.

4 Retrieval turns it around

Fetch only the few documents that resemble the question, and send just those. Cheaper, shorter and more accurate.

RAG is not a bigger prompt — it is a smaller, better-chosen one.

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TOPIC 3 · RAG

The RAG Pipeline — Ingest Once, Retrieve Every Question

Two separate workflows. You run ingestion once; retrieval runs on every single question.

Collect the source documents ▶ **Split** into chunks ▶ **Embed** each chunk into a vector ▶ **Store** in the vector store ▶ **Retrieve** the nearest chunks ▶ **Answer** from what came back

1 Ingestion — run by hand

Lab 5's first workflow: list 20 brochures on Drive, download, split, embed, insert. A Manual Trigger, run once.

2 Retrieval — runs per question

Lab 5's second workflow: embed the question with the SAME model, find the nearest brochures, answer only from those.

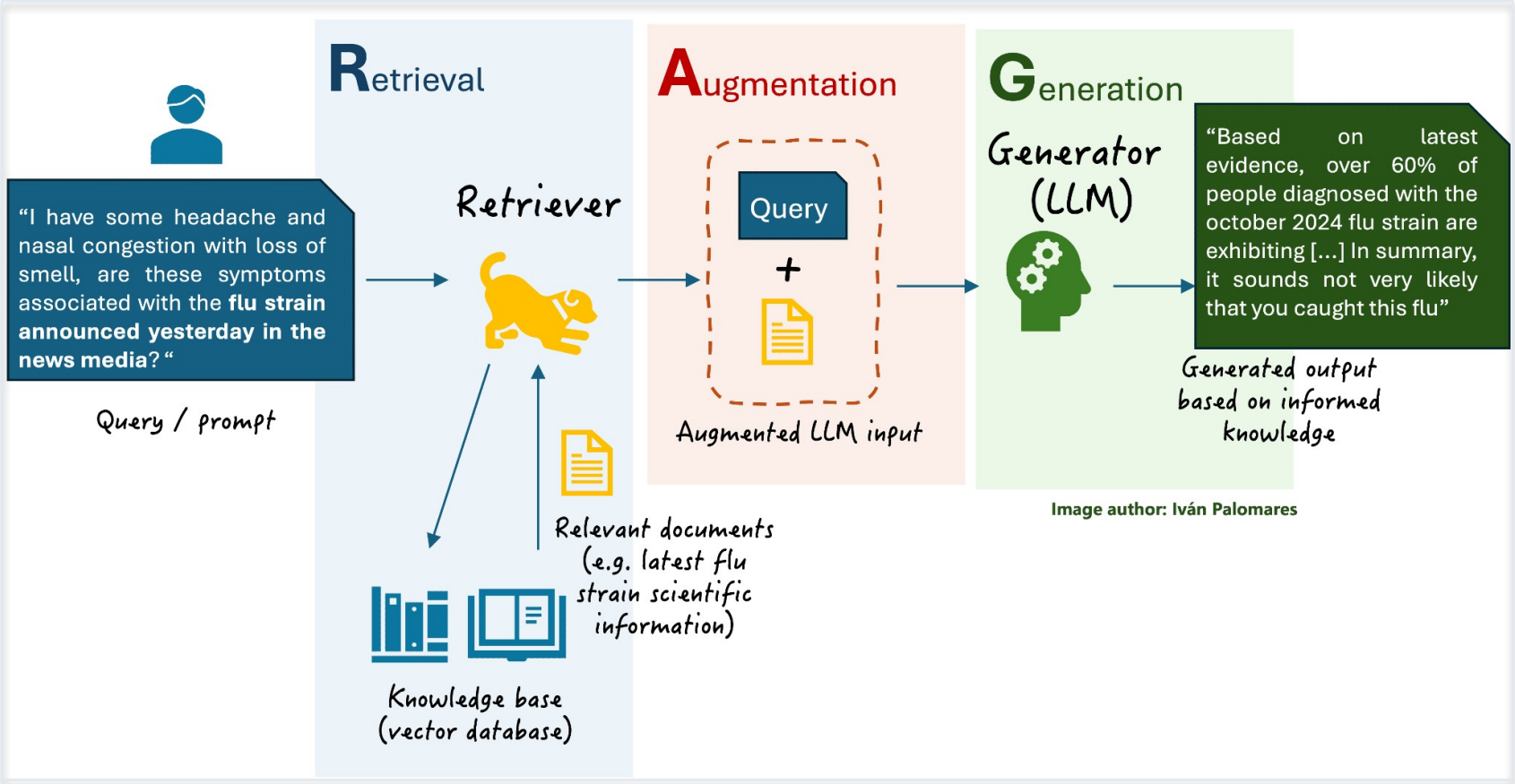
3 The models must match

A question embedded by one model cannot be compared against documents embedded by another. It fails silently, returning nonsense.

Ingest once, retrieve thousands of times — and use the identical embedding model on both sides.

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How RAG Works



Retrieval finds the documents that resemble the question · Augmentation adds them to the prompt · Generation produces an answer grounded in what was retrieved

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TOPIC 3 · RAG

Chunking — the Biggest Lever on RAG Quality

Most RAG projects fail quietly at chunking, not at the model.

1 The rule

The right chunk is the smallest piece of text that still answers a question on its own.

2 Too small

A brochure sliced into thirds separates the course name from its fee. Retrieval then finds one without the other.

3 Too large

Irrelevant text crowds the prompt and dilutes what actually matters.

4 Overlap

A small overlap between chunks stops a sentence being cut in half at the boundary.

5 In Lab 5

Chunk size 4000 with overlap 0 — one brochure becomes exactly one vector, so the fee is never separated from the course.

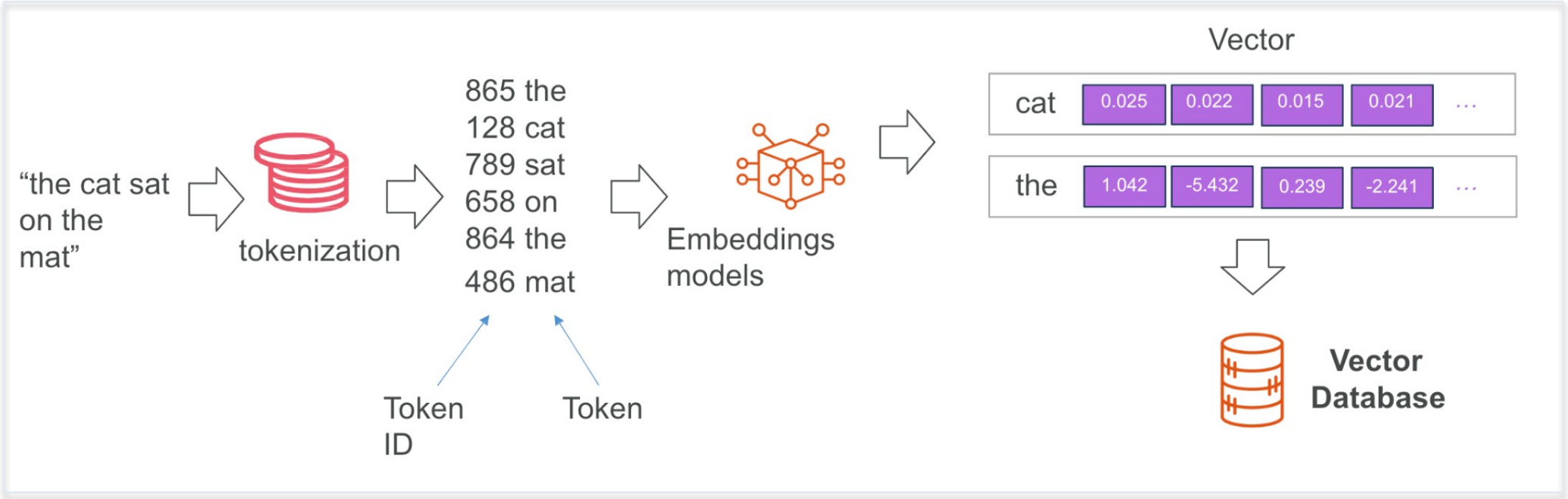
6 topK

How many chunks come back. Too few misses the answer; too many dilutes the prompt with noise.

Tune the chunking before you blame the model — that is where the answer was lost.

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Tokenization and Text Embedding



Text is split into tokens, then an embedding model turns it into a vector — a list of numbers that places similar meanings close together

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Choosing a Vector Store

Where the embeddings live, and whether they are still there tomorrow.

SIMPLE VECTOR STORE — built in

- No account, no API key, no index to create.
- Running RAG within ten minutes — this is what Lab 5 uses.
- WIPED whenever n8n restarts.
- Right for: learning, demos and prototypes.

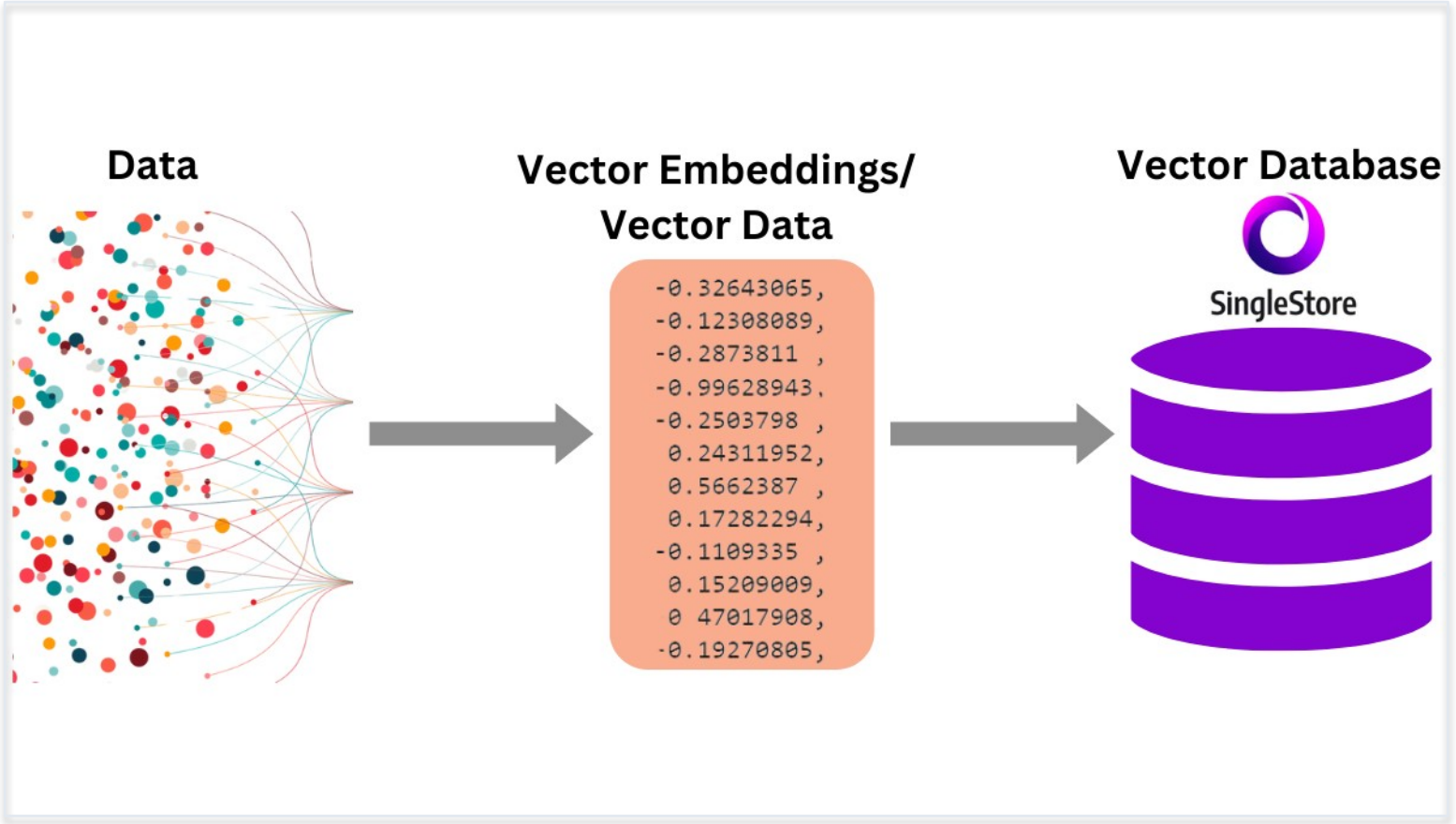
A REAL VECTOR DATABASE

- Pinecone, Supabase (pgvector), Qdrant, Weaviate.
- An account, an index and a matching dimension to set up.
- Persists, scales, and is shared across n8n instances.
- Right for: anything that goes into production.

Prototype on the in-memory store; move to a hosted database the moment anyone else depends on the answers.

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Vector Databases



A vector store holds the embeddings and finds the nearest neighbours to a question, fast

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TOPIC 3 · RAG

Grounding an Agent Against Hallucination

An ungrounded model asked for a price will produce one. It will be plausible, it will be wrong, and the customer will quote it back to you.

1 Always call the tool first

Instruct the agent to search the documents before answering any factual question at all.

2 Answer only from what came back

Every factual claim must trace to a retrieved document, not to the model's general knowledge.

3 Give it permission to fail

'I don't have that in our course information' is a far better answer than a confident invention.

4 Name what it must never invent

A fee, a date, a duration, a course code, an instructor's name, a discount.

5 Hide the machinery

Never mention the vector store or that it is searching documents — the customer does not care.

6 Test the traps

Ask about a course you do not run, a person who does not exist, and a discount never offered.

Grounding is an instruction problem as much as a retrieval one — retrieval finds it, the prompt makes it stick.

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Lab 5 — RAG Customer Care Chatbot — Ingestion Workflow

Cook & Bake Academy's customer care team retypes the same answers all day, and every answer is already written in the course brochures.

Manual Trigger
you run ingestion once

▶

Drive — list + download
20 brochures

▶

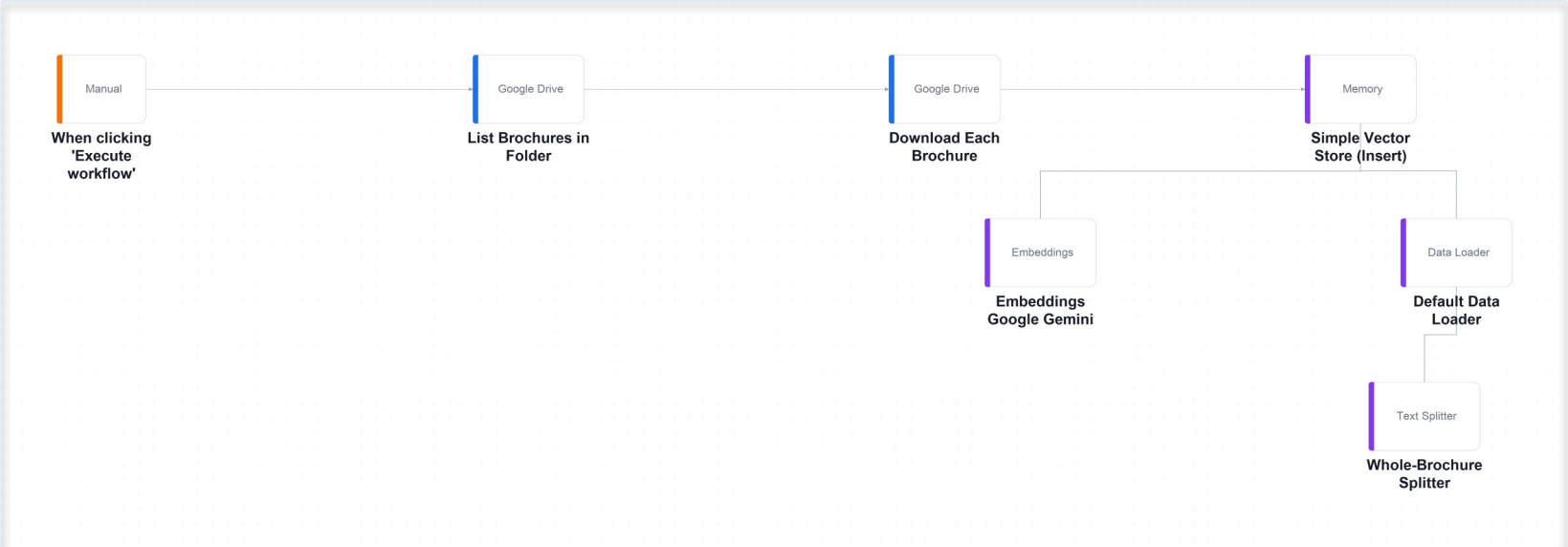
Splitter
1 brochure = 1 chunk

▶

Embeddings
3072-dim vectors

▶

Vector Store — Insert
stored, ready to search



The ingestion workflow on the n8n canvas — run ONCE by hand. Twenty brochures are listed, downloaded, split, embedded and inserted into the vector store.

LAB 5

RAG Customer Care Chatbot — Ingestion Workflow

Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond · full click-by-click steps in the Learner Guide

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Lab 5 — RAG Customer Care Chatbot — Answering Workflow

The answering workflow on the n8n canvas — runs on EVERY question. The Query Data Tool and its embedding model hang below the agent.

Webhook — POST
a visitor asks a question

▶

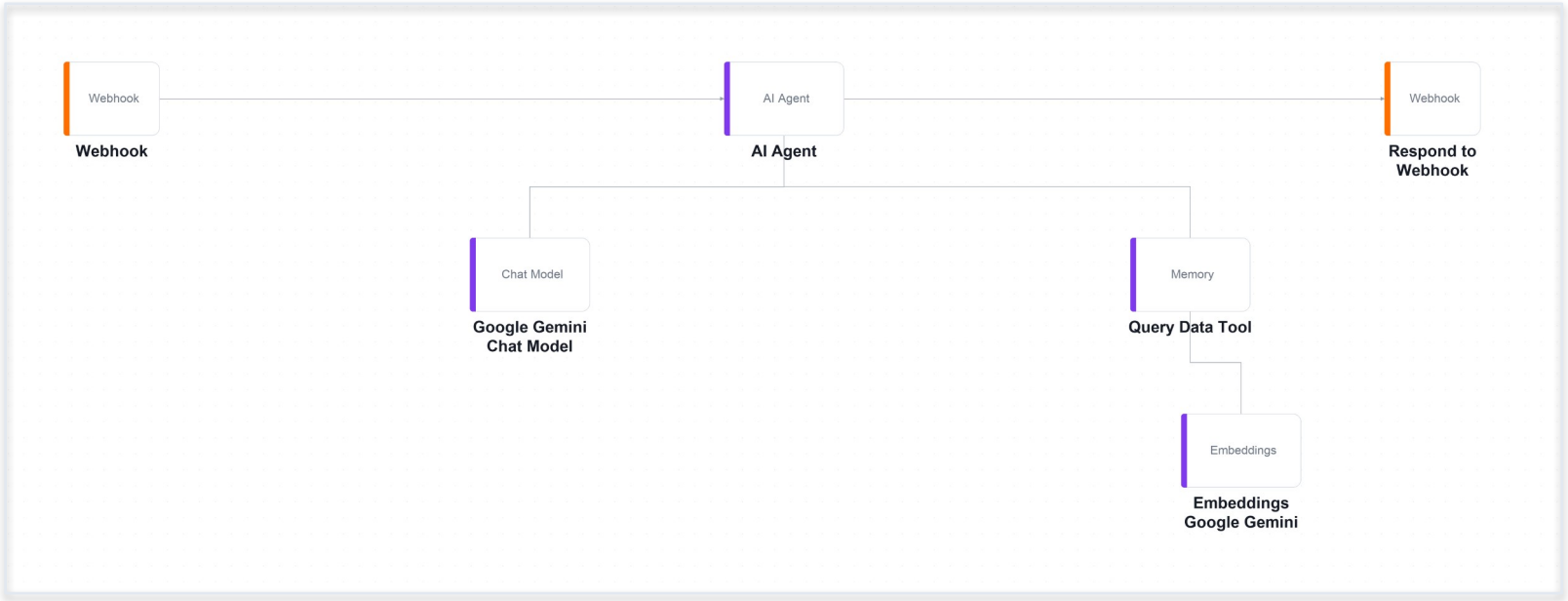
AI Agent
+ Query Data Tool

▶

Retrieve
nearest brochures only

▶

Respond
answer grounded in those



LAB 5

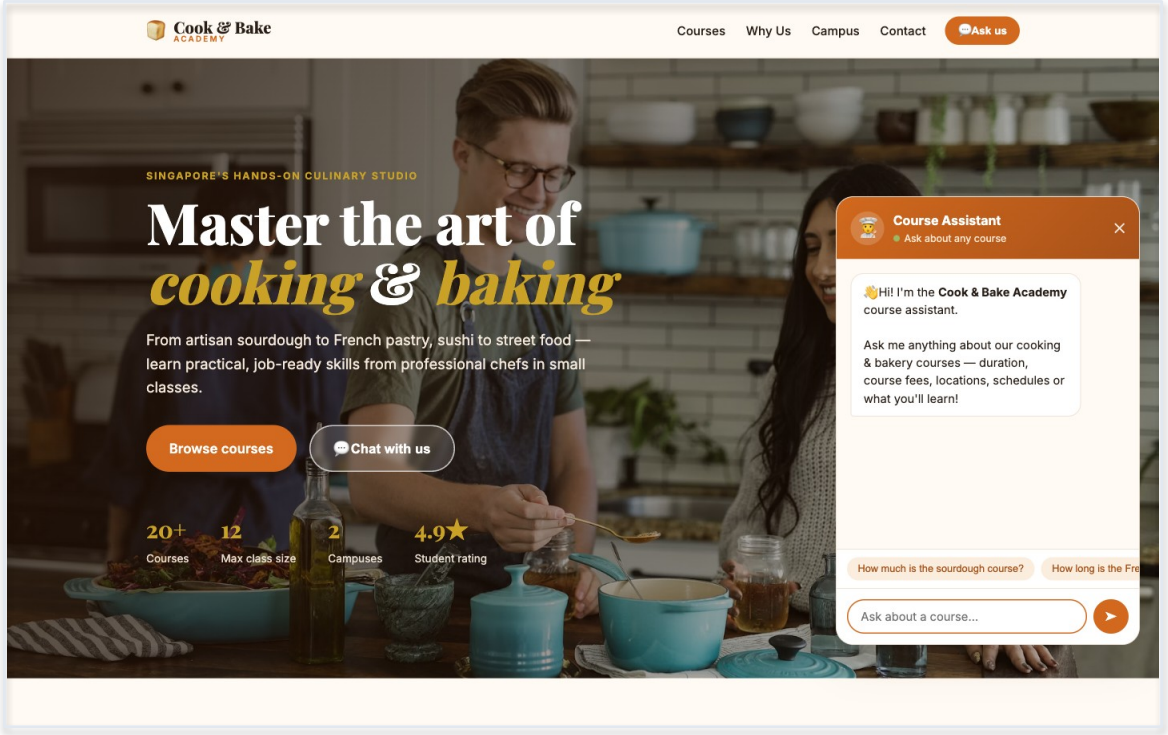
RAG Customer Care Chatbot — Answering Workflow

Ingestion: Drive → Split → Embed → Vector Store · Answering: Webhook → Agent + Retriever → Respond · full click-by-click steps in the Learner Guide

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Lab 5 — RAG Customer Care Chatbot — What You'll Build

- The Cook & Bake Academy site with the course assistant open. Every answer it gives is grounded in the 20 brochures you ingested into the vector store.
- RAG retrieves the few documents that resemble the question instead of pasting everything into the prompt.
- An embedding places similar meanings close together, so retrieval works even when no words match.



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LAB 5 · VERIFY AND DEBRIEF

Lab 5 — RAG Customer Care Chatbot

TC1 must quote BAK-101's exact fee from the brochure. Then run the three hallucination probes: TC7 (a course that does not exist), TC8 (an instructor name no brochure contains) and TC9 (a discount that was never offered). Any confident answer to these is a failure, however fluent.

1 What RAG does

RAG retrieves the few documents that resemble the question instead of pasting everything into the prompt.

2 Embeddings

An embedding places similar meanings close together, so retrieval works even when no words match.

3 Chunking

Chunking is the single biggest lever on RAG quality — the right chunk answers a question on its own.

4 Match model and key

The ingestion and retrieval embedding models must be identical, and the memory keys must match.

5 In-memory is temporary

n8n's Simple Vector Store is wiped when n8n restarts — fine for learning, unacceptable in production.

Lab 5 — you can now deploy a RAG application that grounds an agent in the organisation's own documents.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · n8n and AI agents

Built a trigger-and-action workflow and explained how an LLM agent differs from generative AI.

2

LO2 · Webhook-triggered agents

Put an AI agent behind a real website, with tools, structured output and an audit log.

3

LO3 · Agentic chatbots

Designed a chatbot that collects contact details and refuses to give regulated advice.

4

LO4 · Human in the loop

Paused a workflow on an approval gate so no drafted communication was sent unsupervised.

5

LO5 · RAG

Ingested 20 documents into a vector store and grounded a chatbot so it stopped inventing answers.

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NEXT STEPS

Taking This Back to Work

Six moves that turn what you built this week into something your organisation actually runs.

1 Start with one painful process

Pick something repetitive, high-volume and low-risk. Automate that first.

2 Map it before you build it

Draw the trigger, the decisions and the actions on paper. The canvas comes second.

3 Decide where the human belongs

Anything irreversible, regulated or expensive keeps an approval gate.

4 Ground the agent in your own data

RAG over your policies and documents beats a general model every time.

5 Log everything from day one

You will need the audit trail long before you think you will.

6 Iterate with real cases

Test with the questions you fear, not the ones you hope for.

The next process you automate is the one you were dreading this morning.

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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

Support

1

Email

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Telephone

+65 6100 0613

3

Website

www.tertiarycourses.com.sg

4

LMS / TMS portal

lms-tms.tertiaryinfotech.com

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Assessment

1

Written Assessment (SAQ)

1 hour. Short-answer questions covering the course knowledge.

2

Practical Performance (PP)

1 hour. Hands-on n8n build tasks drawn from the five labs.

3

Open book

Slides, Learner Guide and approved materials only.

4

Submit on the LMS

Upload your completed answers at lms-tms.tertiaryinfotech.com.

5

Digital attendance

Remember to take the Assessment digital attendance (TRAQOM).

6

Appeals

An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

It is mandatory to take all three digital attendances for WSQ-funded courses.

2

The QR code

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan it with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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HAPPY AUTOMATING

Thank You!

Keep your n8n instance running and keep building. The next process you automate is the one you were dreading this morning.

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